Index

Sl.No.	Chapter Name	Page No.
1.	Abstract	3
2.	Introduction	3
3.	Literature Review / Background Study	3
4.	Problem Statement	3
5.	System Requirements	4
6.	System Design	4
7.	Implementation	5-8
8.	Results and Output	9-10
9.	Discussion and Analysis	11
10.	Applications and Future Scope	11
11.	Conclusion	11-12
12.	References	12

1.ABSTRACT:

The Fake News Text Analyzer is a simple program developed using the functional programming language Haskell. The main goal of this project is to analyze a given news article and determine whether it may contain fake or misleading information. The program reads a news text file and checks for certain keywords that are commonly used in fake news articles such as “shocking”, “breaking”, or “secret”. Based on the number of such words found in the article, the program calculates a fake score and classifies the news as likely fake or possibly real. This project demonstrates how functional programming concepts like list processing, higher-order functions, and file handling can be applied to perform simple text analysis.

2.INTRODUCTION:

Fake news has become a major issue in the digital world because information spreads very quickly through social media and online platforms. Many news articles use sensational words and exaggerated language to attract readers even when the information is not accurate. Because of this problem, it is useful to build simple tools that can analyze news content and identify suspicious patterns.

In this project, a Fake News Text Analyzer is implemented using Haskell. The program reads a news article from a text file, processes the text, and checks for fake news indicators using keyword matching. The result helps in identifying whether the article may be fake or not.

3. Literature Review / Background Study:

Many researchers and developers have worked on detecting fake news using different techniques such as machine learning, natural language processing, and statistical analysis. These systems analyze patterns in news articles, social media posts, and user behavior.

However, such systems are often complex and require large datasets. For learning purposes, a simpler approach can be used by analyzing text and checking for common fake news keywords. Functional programming languages like Haskell provide powerful tools for processing lists and strings, which makes them suitable for this type of analysis.

4. Problem Statement:

The main problem addressed in this project is identifying whether a given news article may contain fake or misleading information.

Fake news articles often use dramatic words and emotional language to attract attention. By detecting these keywords and analyzing the text content, the program attempts to estimate the likelihood of the news being fake.

The goal of the project is to create a simple text analyzer that reads a news file, searches for fake news indicators, and displays a classification result.

5. System Requirements:

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Standard keyboard and display

Software Requirements

- Windows operating system
- PowerShell terminal
- Glasgow Haskell Compiler (GHC)
- GHCi interactive environment
- Text editor (Notepad or VS Code)

6. System Design:

The system is designed as a simple command-line application.

First, the user provides the name of a news article text file. The program reads the content of the file and converts all text to lowercase for easier comparison. It then splits the text into individual words.

Next, the program compares these words with a predefined list of fake news keywords stored in another file. The number of matching keywords is counted. Based on this count, the program calculates a fake score and displays the result.

The system mainly consists of three parts:

1. Input processing
2. Keyword analysis
3. Result generation

7.IMPLEMENTATION:

The project is implemented using the Haskell programming language. The program reads the news article from a file using file input functions. The text is then converted to lowercase and split into words.

Higher-order functions such as map and filter are used to process the list of words. The program checks whether each word exists in the list of fake news keywords. If matches are found, they are counted and used to calculate a fake score.

If the score exceeds a certain threshold, the article is classified as likely fake news. Otherwise, it is considered possibly real.

CODE:

```haskell
import System.IO
import System.Process
import Data.Char
import Data.List

-----------------------------------------------
-- Keyword Lists
-----------------------------------------------

fakeKeywords :: [String]
fakeKeywords =
  ["breaking","shocking","secret","conspiracy","fraud","scandal",
   "leaked","exclusive","urgent","miracle","exposed","rumor",
   "panic","danger","bombshell","sensational"]

clickbaitWords :: [String]
clickbaitWords =
  ["unbelievable","amazing","incredible","you","won't","believe",
   "secret","revealed","exposed","truth","shocking"]

-----------------------------------------------
-- Text Cleaning
-----------------------------------------------

cleanText :: String -> String
cleanText = map toLower . map replace
  where
    replace c = if isAlpha c then c else ' '

-----------------------------------------------
-- Tokenization
-----------------------------------------------

tokenize :: String -> [String]
tokenize = words . cleanText
```

```haskell
-- Keyword Counter
-----------------------------------------------

countKeywords :: [String] -> [String] -> Int
countKeywords keywords ws =
  length (filter (`elem` keywords) ws)

-----------------------------------------------
-- Capital Letter Ratio
-----------------------------------------------

capitalRatio :: String -> Double
capitalRatio text =
  let caps = length (filter isUpper text)
      letters = length (filter isAlpha text)
  in if letters == 0
     then 0
     else fromIntegral caps / fromIntegral letters * 100

-----------------------------------------------
-- Exclamation Count
-----------------------------------------------

exclamationScore :: String -> Double
exclamationScore text =
  let exc = length (filter (=='!') text)
  in fromIntegral exc

-----------------------------------------------
-- Fake Score Calculation
-----------------------------------------------

fakeScore :: Double -> Double -> Double -> Double -> Double
fakeScore kw cb cap exc =
  (kw * 0.4) +
```

```haskell
fakeScore kw cb cap exc =
  (kw * 0.4) +
  (cb * 0.2) +
  (cap * 0.2) +
  (exc * 0.2)

-----------------------------------------------
-- Classification
-----------------------------------------------

classify :: Double -> String
classify score
  | score > 25 = "HIGH PROBABILITY OF FAKE NEWS"
  | score > 10 = "SUSPICIOUS NEWS"
  | otherwise  = "LIKELY REAL NEWS"

-----------------------------------------------
-- Main Program
-----------------------------------------------

main :: IO ()
main = do

  putStrLn "-------------------------------------"
  putStrLn "FAKE NEWS DETECTION SYSTEM"
  putStrLn "Haskell Functional Analyzer + AI"
  putStrLn "-------------------------------------"

  putStrLn "Enter news text file:"
  file <- getLine

  content <- readFile file

  let tokens = tokenize content
  let totalWords = length tokens
```

```haskell
-- feature extraction

let kwMatches = countKeywords fakeKeywords tokens
let cbMatches = countKeywords clickbaitWords tokens
let capScore = capitalRatio content
let excScore = exclamationScore content

let kwScore = fromIntegral kwMatches / fromIntegral totalWords * 100
let cbScore = fromIntegral cbMatches / fromIntegral totalWords * 100

let final = fakeScore kwScore cbScore capScore excScore
let result = classify final

-----------------------------------------------
-- Output
-----------------------------------------------

putStrLn ""
putStrLn "------ HASKELL ANALYSIS ------"

putStrLn ("Total Words: " ++ show totalWords)

putStrLn ("Fake Keyword Matches: " ++ show kwMatches)
putStrLn ("Clickbait Word Matches: " ++ show cbMatches)

putStrLn ("Keyword Score: " ++ show kwScore ++ "%")
putStrLn ("Clickbait Score: " ++ show cbScore ++ "%")
putStrLn ("Capital Letter Score: " ++ show capScore ++ "%")
putStrLn ("Exclamation Score: " ++ show excScore)

putStrLn ("Combined Fake Score: " ++ show final ++ "%")

putStrLn ("Haskell Result: " ++ result)

```

```haskell
-----------------------------------------------
-- AI Verification
-----------------------------------------------

putStrLn ""
putStrLn "------ AI VERIFICATION ------"

callCommand ("python predict_news.py " ++ file)
```

8. Results and Output:

The program successfully analyzes a news article and produces an output based on the number of fake news keywords found.

Samplefakenews.txt

```
Breaking Secret Conspiracy Exposed!
Breaking shocking news has exposed a secret conspiracy that many officials tried to hide.
According to viral reports circulating online, a hidden plan has been revealed that could change
everything. Sources claim the truth was deliberately covered up for years.
The unbelievable scandal has created panic among people after leaked documents suggested that
a forbidden experiment was secretly conducted in several locations. Experts are warning that the
situation could be dangerous if the full truth is not revealed.
Many sensational headlines claim the exposed conspiracy involves illegal activities and fraudulent
operations carried out in secrecy. Viral stories and dramatic reports continue spreading quickly
across social media platforms.
Authorities have not confirmed these shocking claims, but rumors and alarming stories continue to
circulate. Several dramatic warnings and urgent alerts have appeared online, increasing public
concern and confusion
```

```
Enter news text file:
fake.txt

------ HASKELL ANALYSIS ------
Total Words: 140
Fake Keyword Matches: 17
Clickbait Word Matches: 12
Keyword Score: 12.142857142857142%
Clickbait Score: 8.571428571428571%
Capital Letter Score: 1.542111506524318%
Exclamation Score: 1.0
Combined Fake Score: 7.079850872733436%
Haskell Result: LIKELY REAL NEWS

------ AI VERIFICATION ------
----------------------------------
FAKE NEWS ANALYSIS REPORT
----------------------------------
Prediction : FAKE NEWS
Confidence : 99.1 %
```

Sampletruenews.txt

```
Renewable Energy Expansion in India
India has made significant progress in expanding renewable energy capacity as part of its
long term strategy to reduce carbon emissions and improve energy security. According to recent
reports, non fossil fuel sources such as solar, wind, hydro, and nuclear power now account for
nearly half of the country's installed electricity capacity.
Government officials stated that the country is continuing to invest heavily in renewable
infrastructure, including large solar parks, wind farms, and battery storage systems. Experts say
that these investments are important for meeting the nation's goal of achieving 500 gigawatts of
non fossil fuel power capacity by 2030.
The expansion of renewable energy is also expected to create new employment opportunities in
engineering, construction, and maintenance sectors. Analysts note that falling costs of solar panels
and wind technology have made clean energy more competitive compared to traditional fossil fuels.
Energy researchers believe that increasing renewable capacity will help reduce dependence on
imported fuels and improve environmental sustainability. Several new projects are already under
development across different states, and the government plans to continue supporting clean energy
innovation and research in the coming years.
```

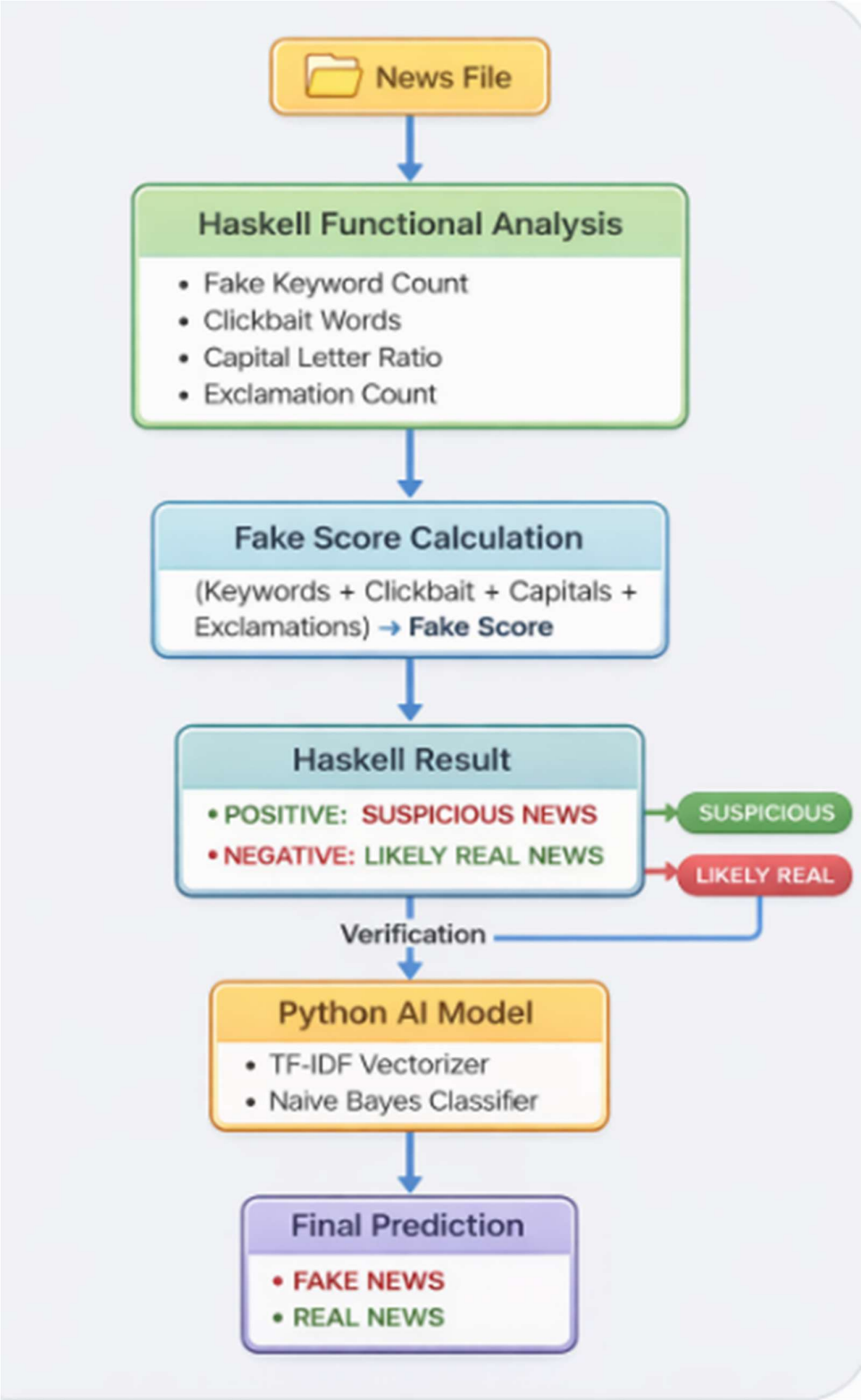
```
Enter news text file:
true.txt

------ HASKELL ANALYSIS ------
Total Words: 188
Fake Keyword Matches: 0
Clickbait Word Matches: 1
Keyword Score: 0.0%
Clickbait Score: 0.5319148936170213%
Capital Letter Score: 1.6143497757847534%
Exclamation Score: 0.0
Combined Fake Score: 0.42925293388035496%
Haskell Result: LIKELY REAL NEWS

------ AI VERIFICATION ------
----------------------------------
FAKE NEWS ANALYSIS REPORT
----------------------------------
Prediction : FAKE NEWS
Confidence : 59.56 %
----------------------------------
```

This output shows how the program calculates the fake score and displays the classification result.

FLOW CHART


News File
Haskell Functional Analysis
• Fake Keyword Count
• Clickbait Words
• Capital Letter Ratio
• Exclamation Count
Fake Score Calculation
(Keywords + Clickbait + Capitals + Exclamations) → Fake Score
Haskell Result
• POSITIVE: SUSPICIOUS NEWS
• NEGATIVE: LIKELY REAL NEWS
SUSPICIOUS
LIKELY REAL
Verification
Python AI Model
• TF-IDF Vectorizer
• Naive Bayes Classifier
Final Prediction
• FAKE NEWS
• REAL NEWS

9. Discussion and Analysis:

The project demonstrates how functional programming can be used to perform basic text analysis. The keyword matching approach is simple but effective for detecting certain patterns in fake news articles.

However, this approach has some limitations. Not all fake news articles contain obvious keywords, and some real news articles may also use similar words. Therefore, the accuracy of this system may not always be perfect.

Despite these limitations, the project provides a good example of applying functional programming techniques to solve a real-world problem.

10. Applications and Future Scope:

This type of system can be used as a basic tool for analyzing news articles before sharing them online. It can help users identify potentially misleading information.

In the future, the system can be improved by:

- Using larger keyword datasets
- Applying natural language processing techniques
- Integrating machine learning models
- Creating a web interface for easier use

These improvements could increase the accuracy and usefulness of the fake news detection system.

11. Conclusion:

The Fake News Text Analyzer project shows how functional programming concepts can be applied to analyze textual data. The system reads a news article, searches for fake news keywords, and calculates a score to classify the article.

Although the current implementation is simple, it demonstrates the potential of functional programming for building text analysis tools. The project also provides practical experience with file handling, list processing, and higher-order functions in Haskell.

12. References:

Haskell Programming Documentation

Online tutorials on functional programming

Articles related to fake news detection

Programming examples from Haskell learning resources